

ST-MRAM Fundamentals

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Abstract— Magnetoresistive Random Access Memory (MRAM) technology emerged from research and development into volume production within the last decade in the form of Toggle MRAM. The latest Magnetic Tunnel Junction (MTJ) based memory technology, Spin-Torque MRAM, has reached the level of customer sampling, offering higher density and bandwidth. Spin-Torque MRAM enables new applications, offers a wide range of features for use in embedded memory, and has the potential to extend to technology nodes beyond the capability of DRAM. This paper describes the devices, fundamental circuit challenges, and applications of this evolving MTJ based memory.

I. INTRODUCTION.

Magnetic materials have been used in data storage elements since the early days of transistor based memory devices [1]. In the early 1990s, research into MRAM began using Giant Magnetoresistive (GMR) devices, with the potential of yielding a fast, unlimited endurance, non-volatile memory [2]. The pursuit of MRAM shifted in focus from GMR to Magnetic Tunnel Junction (MTJ) based devices in the late 1990s [3, 4]. These early MRAM approaches switch between two stable states using magnetic fields generated by currents flowing through conductors in close proximity to the storage elements (Fig. 1a). Of these field based MRAM approaches, the most robust and the basis for today's volume production devices is Toggle MRAM [5]. Toggle MRAM, available in densities up to 16Mb, has been adopted in a wide range of applications that previously required SRAM with either a battery backup supply or infrequent backup of the SRAM to conventional non-volatile devices to achieve fast non-volatile storage with unlimited endurance. Generating the field to switch between states in these MRAM devices, however, requires a significant level of current which does not reduce as the MTJ scales to smaller geometries. To address this challenge to scaling of field based MRAM devices, a new approach to switching the state of the MTJ has evolved called Spin-Torque MRAM (ST-MRAM) [6]. ST-MRAM switches the MTJ between two stable states by flowing current directly through the MTJ (Fig 1b). Development of ST-MRAM over the last several years has resulted in fully functional 64Mb DDR3 ST-MRAM devices [7, 8]. Scaling of ST-MRAM improves greatly over that of field based MRAM since the current required for switching reduces as the area of the MTJ

is scaled to smaller geometries. This lower switching current enables not only scaling to increased densities, but also enables sustained high-bandwidth write operations at DDR3 performance levels and beyond. Maintaining a required level of data retention, however, is a challenge for ST-MRAM as the MTJ is scaled. The next evolution of MTJ based memory uses perpendicular magnetic anisotropy (PMA) to achieve improved data retention at lower switching currents and has been considered a prime candidate for high density memory at nodes below 20nm [9].

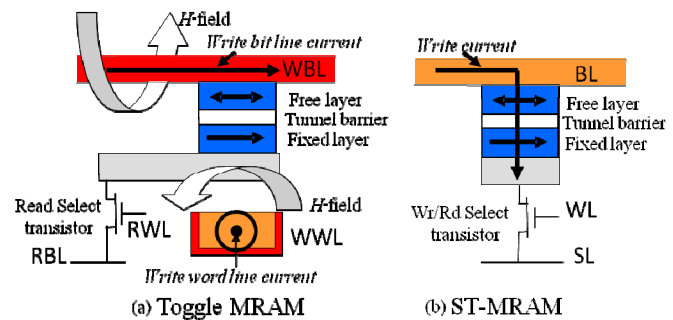


Fig. 1. (a) Field switching MRAM; (b) Spin-Torque switching MRAM

II. MAGNETIC TUNNEL JUNCTION STATES

A magnetic tunnel junction is constructed using a reference layer, typically with a fixed magnetic polarization, a storage layer that can be switched between two stable magnetic polarization states, and a tunnel barrier between the two magnetic layers. The resistance states of the MTJ are distinguished by the magnetoresistance effect, wherein the resistance across the tunnel barrier depends on the magnetic polarization of the storage layer relative to the reference layer. As shown in Fig. 2a, when the magnetic polarization of the storage layer is aligned with the magnetic polarization of the reference layer, the tunnel barrier exhibits a lower tunneling resistance and when the polarization of the storage layer is opposite that of the reference layer, the tunnel barrier exhibits a higher tunneling resistance.

The Magnetoresistance Ratio (MR) is defined as the percentage increase in resistance from the low resistance state to the high resistance state. MR, which is typically quoted using a very low bias measurement, decreases as the magnitude of the voltage across the MTJ increases (Fig. 1b). MR also decreases with increasing temperature.

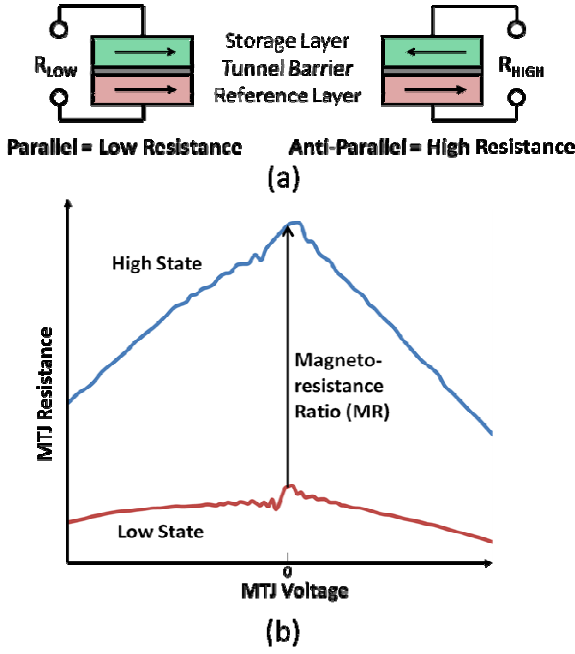


Fig. 2. (a) MTJ resistance states; (b) Resistance vs bias voltage

III. FIELD SWITCHING MRAM

MRAM products available in the market today use magnetic field to switch the storage layer between the two states. Initial approaches to switching MTJ devices apply a hard axis field along a row of MTJ elements and easy axis fields along columns of MTJ elements, switching the MTJ elements at the intersection of the row and columns when sufficient field is applied to each axis. With this initial approach, the state written to the MTJ depends on the polarity of the easy axis field. One significant challenge to this approach is that half selected devices along a row or column which are exposed to a single axis of applied field can be disturbed with excess current in one axis. Achieving a reliable window of switching functionality with this initial approach requires a very tight distribution of switching thresholds for the MTJ elements in an array.

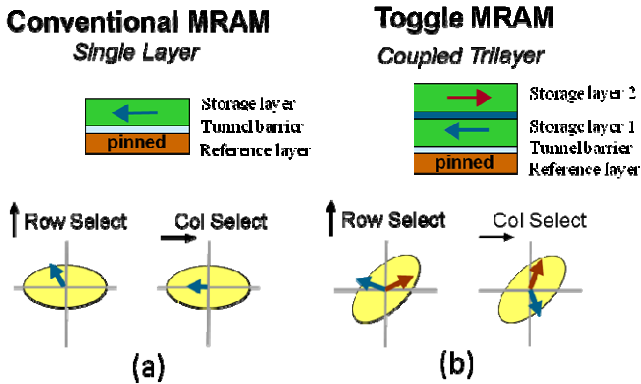


Fig. 3. (a) Basic MTJ and (b) Toggle MTJ, showing storage layer response to half selects on each axis

Toggle MRAM was developed to address this challenge using a different MTJ stack and an offset timing sequence on

the programming lines. The Toggle MTJ, as seen in Fig. 3, adds a second storage layer to the stack, coupled to the first storage layer with a synthetic anti-ferromagnetic coupling layer that acts to keep the magnetic polarization of the two storage layers opposite to each other. The Toggle MTJ is oriented at an angle relative to the programming lines, which, with the stability added by the coupling layer, provides significant immunity to half select disturbs.

Switching the Toggle MTJ between states requires an offset timing sequence applied to the programming lines as shown in Fig. 4. The sequence applied to the program lines toggles the state from high to low resistance or from low to high resistance. Therefore, in order to set the Toggle MTJ to desired the state, the initial state of the MTJ must be read and a toggle sequence applied only if necessary.

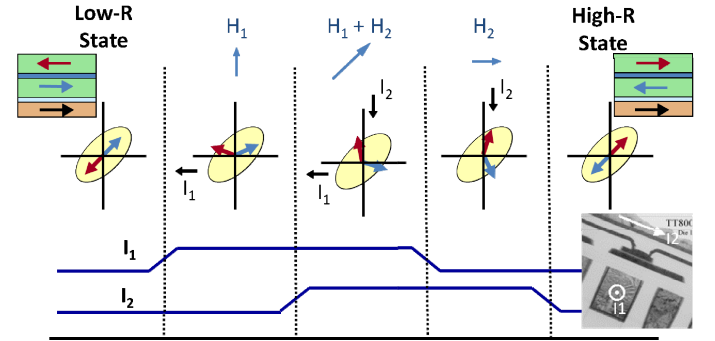


Fig. 4. Toggle MRAM switching sequence

The improved switching window of Toggle MRAM is demonstrated in Fig. 5. The initial conventional approach achieves only a very narrow window of operation with sufficient current applied to switch all the MTJ elements in an array without excess current on either axis causing half select disturbs. In contrast, Toggle MRAM is tolerant of significant overdrive on each axis beyond the point where all MTJ elements are switched. There is, however, an upper bound to the applied current for Toggle MRAM beyond which the coupling between the storage layers is saturated and switching is not guaranteed. This saturation level decreases with increasing temperature, reducing the operating window slightly.

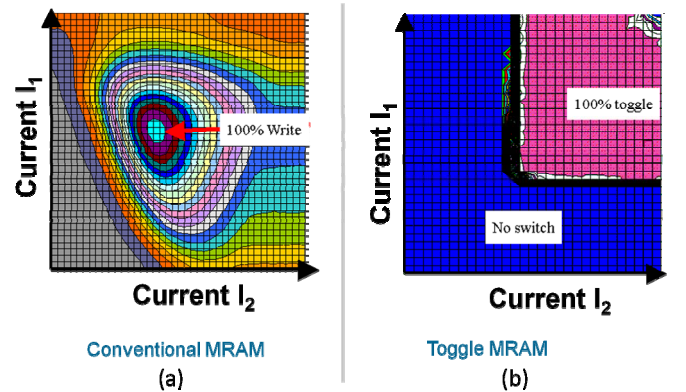


Fig. 5. Switching window for (a) initial conventional MRAM, and (b) Toggle MRAM

To avoid the saturation current level at high temperature

while applying sufficient current to switch the entire population of bits reliably across temperature, adjustment of the applied current level across temperature is typically required. Figure 5 shows an example of the average switching and saturation fields for each axis over temperature.

Toggle MRAM technology has enabled robust and reliable commercially available products, however, there are challenges in scaling the technology to support higher densities. The field magnitude required for field switching MRAM generally increases for smaller MTJ. The required program line current can be maintained by moving the metal lines closer to the MTJ, however, any significant reduction in current is difficult to achieve. Without significant reduction in switching current, which can be well over 10mA per axis, significant reductions are not likely in the supply voltage level required to sustain the IR drop along the program lines, the area of the circuitry controlling the switching current, or the power required to achieve an increased write bandwidth.

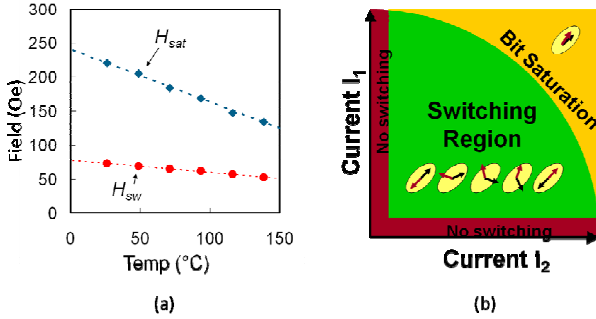


Fig. 6. (a) Average switching and saturation field; (b) Switching and saturation regions

IV. SPIN-TORQUE MRAM

Spin-Torque (Transfer) MRAM (ST-MRAM, or STT-MRAM) switches the state of the MTJ by flowing a sufficient density of current (J_c) across the tunnel barrier. When switching from the high resistance (anti-parallel) to the low resistance (parallel) state, current flows from the free storage layer to the fixed reference layer. As electrons move through the fixed layer, their spin is polarized to match the fixed layer. When a sufficient density of polarized electron flow is achieved, the free layer changes state to align with the fixed layer. Switching from the low resistance (parallel) to the high resistance (anti-parallel) state is accomplished by applying current in the opposite direction, with current flowing from the fixed reference layer to the free storage layer. In this switching direction, electrons with spin opposite the fixed layer are reflected back to the free layer causing the free layer to switch to the state opposite to the fixed layer.

One key component of the MTJ material stack for ST-MRAM is the use of Magnesium Oxide for the tunnel barrier rather than the Aluminum oxide used in Toggle MRAM today. As shown in Figure 7, a significantly lower resistance-area product (RA) can be achieved with Magnesium Oxide than typically achieved with Aluminum oxide. This lower RA is essential to achieving the required current density for Spin-Torque switching at voltages low enough for reliable operation.

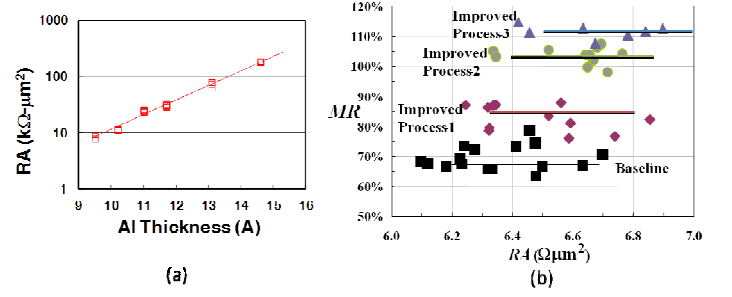


Fig. 7. (a) RA vs Al thickness for Aluminum Oxide; (b) MR vs. RA for Magnesium Oxide

Figure 8 shows an example of how ST-MRAM may scale relative to the current drive of a minimum feature size transistor, such as in a DRAM device [8]. The I_c (MTJ) curve represents the average current required to switch an MTJ and the I_{sw} (array) curve shows that an increased level of current is required to switch the distribution of MTJ in an array.

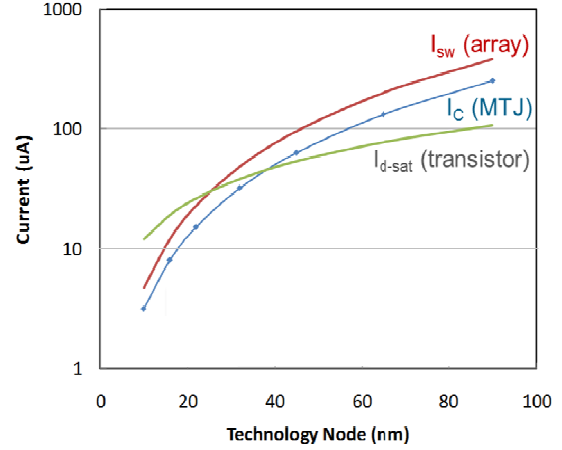


Fig. 8. Example scaling curve for ST-MRAM switching current

Compared to the >10mA of switching current on each axis for Toggle MRAM switching, a reduction to hundreds of microamps to switch an ST-MRAM bitcell enables 90nm products to achieve the DDR3-1600 write bandwidth of 1.6-3.2GB/s chip compared to the 30-60MB/s achieved by existing Toggle MRAM products consuming similar write power. Unlike Toggle MRAM, the required switching current for ST-MRAM is reduced proportional to the area of the MTJ, scaling faster at smaller technology nodes compared to the more linear scaling of the minimum sized transistor drive current. At the scaling rate described in Fig 8, ST-MRAM becomes a viable candidate for DRAM replacement at technology nodes below 20nm. Significant challenges exist to achieving this scaling path, one of which is maintaining a minimum level of data retention as the MTJ is scaled.

Data retention time for the MRAM device is an exponential function of the energy barrier (E_b) to reverse the state. Energy barrier is typically measured in units of E_b/kT , where k is Boltzmann's constant and T is temperature in degrees Kelvin. The average E_b/kT required to achieve 10 year data retention depends strongly on the E_b distribution (sigma) for each MTJ in the array, but can be in the order of 60-80. Figure 9 shows measurements of average E_b/kT for MTJ with different area

and free layer thickness. Scaling of the MTJ to smaller area causes a reduction in data retention which can be compensated for with an increase in the thickness of the free layer. Increasing the free layer thickness, however, can increase the current density (J_c), and therefore voltage, required to switch the MTJ unless other improvements are made to the MTJ that reduce switching voltage without reducing E_b . MTJ using perpendicular magnetic anisotropy (PMA) have been shown to have an increased E_b for a given J_c compared to MTJ with in-plane magnetic anisotropy and therefore are the focus of research for ST-MRAM at advanced process nodes.

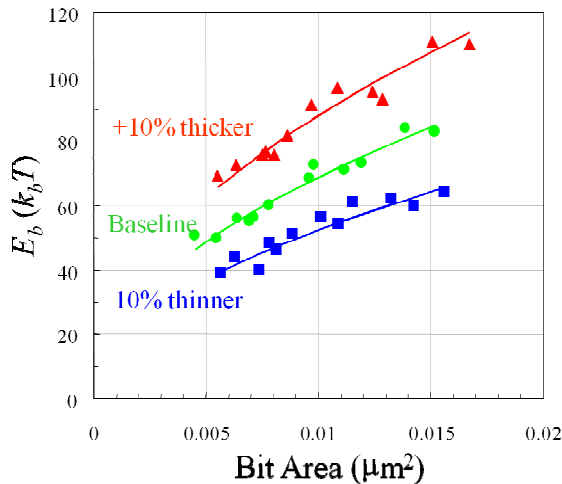


Fig. 9. E_b/kT vs. MTJ area for different relative free layer thicknesses

One key challenge in achieving a reliable ST-MRAM product is applying a sufficiently high write voltage to switch the distribution of MTJ in an array without overstressing the MTJ and limiting the device endurance. Fig. 10 shows an example switching distribution and example breakdown distributions for an array of MTJ [11]. The Time Dependent Dielectric Breakdown (TDDB) failure mechanism requires lower applied voltage levels to achieve higher levels of endurance, which is shown in the shift from the one cycle breakdown curve to the breakdown distribution for many cycles. Sustaining reliable operation over the product lifetime requires good control of the applied write voltage.

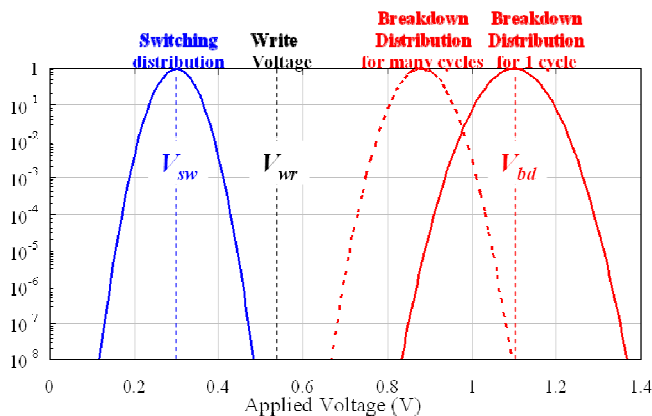


Fig. 10. ST-MRAM applied write voltage window vs. switching and breakdown distributions

Abstract

This chapter covers the entire gamut of MRAM, from the fundamental physics of magnetism, magnetic materials used in MTJ elements, history of MRAM developments, and innovations which were keys to MRAM, to important designing points, MRAM scalability, and expected future evolutions of MRAM.

The chapter is expected to help engineers who are not familiar with MRAM to learn MRAM and develop a deeper understanding of its concepts, so that they can be more creative in their work. The chapter will also help researchers to make innovations for scalability and for memory hierarchy evolution with MRAM.

List of Abbreviations

AMR	Anisotropic Magneto Resistance
CIMR	Current Induced Magnetization Reversal
ECC	Error Correction Code
FeRAM	Ferroelectric RAM
GMR	Giant MagnetoResistance
MRAM	Magnetoresistive Random Access Memory
MTJ	Magnetic Tunnel Junction
NOC	Normally-Off Computing
PRAM	Phase change RAM, also called PCRAM, PCM, or Ovonic Memory
ReRAM	Resistive RAM
Spin-RAM	Alternative term for STT-writing MRAM
TMR	Tunnel MagnetoResistance

Introduction

Figure 1 shows an access-speed vs. density map of existing memories. The ideal memory, a fast and dense nonvolatile memory with unlimited endurance, does not exist. Consequently, most systems use a combination of working memories such as SRAM and DRAM and storage memories such as NAND Flash and HDD as shown in Fig. 2a.

The working memories have fast read/write access speed and unlimited endurance but do not have nonvolatility. The storage memories have nonvolatility and density but do not have fast read/write access speed. They work cooperatively with each other to attain both fast accessibility and nonvolatility of data.

However, there are drawbacks. When such a system is turned on, data, which is located in storages should be copied to the working memories to set up memory systems for usage. The setup is called booting and it takes about a minute in the case of personal computers. When such a system is in use, the working memories consume a lot of energy to keep those data because SRAM leaks current and DRAM needs refresh operation. The battery power consumption is a major problem for mobile systems such as cellular phones.

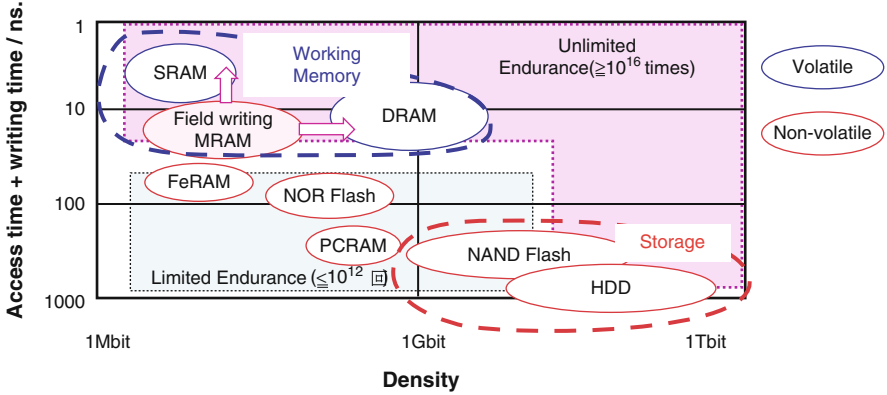


Fig. 1 Access-speed-density mapping of existing memories and targets for MRAM. There are two major markets, work memories and storages. MRAM is a nonvolatile work memory but a commercialized MRAM is neither not very dense nor very fast. Challenges for MRAM are indicated by the two *arrows*, a faster memory like SRAM and a denser memory like DRAM

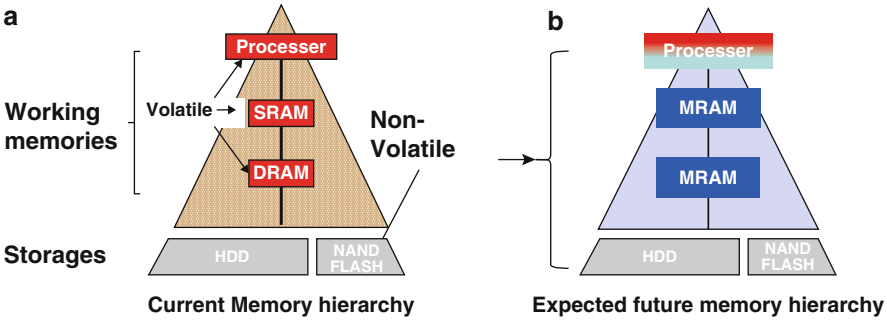


Fig. 2 Memory storage hierarchy used in systems. Since working memories are nonvolatile and storages are slow, a combination of working memories and storage memories are used. MRAM has a potential to replace the working memories

Fast nonvolatile memories with unlimited endurance have a potential to solve these problems. MRAM is the only nonvolatile memory that has relatively fast read/write accessibility and unlimited endurance [1].

Future memory hierarchy is expected as in Fig. 2b. However, an innovation was needed. The commercialized MRAM is not as fast as SRAM, not as dense as DRAM.

Much work has been done on MTJ materials and MRAMs. In particular, spin transfer torque (STT)-writing on MTJs with perpendicular magnetization (simply expressed as P-MTJs) solves most of the problems [2–9]. As a result, it is thought that we are very close to solving the above-mentioned drawbacks of the present SRAM, DRAM/NAND, and HDD memory hierarchy by designing a normally-off memory hierarchy which uses MRAMs.

Definition of MRAM

MRAM is defined as a memory which uses magnetoresistance effect for reading principle regardless of writing principles. There are AMR (Anisotropic Magneto-Resistance), GMR (Giant MagnetoResistance), and TMR (Tunnel MagnetoResistance) for magnetoresistance effect. Commercialized MRAM has used TMR. Elements which have TMR are called MTJs (Magnetic Tunnel Junctions). Various materials for MTJ are presented in Part III, Volume 1.

MRAM, which uses STT writing as a writing principle, is called STT-MRAM, Spin RAM, or STT-RAM.

Figure 3 shows a field writing MRAM and the schematic drawing of its array. The unit cell consists of a selection transistor and a magnetoresistive element.

Digital data is stored in the MTJ, as magnetization directions of a storage layer, rightward or leftward in case of in-plane MTJ, and upward or downward in case of perpendicular MTJ as shown in Fig. 4.

Current MRAMs use MTJs as magnetoresistive elements as shown in Fig. 4. MTJs consist of at least two magnetic layers and a tunneling barrier sandwiched by them. One of the magnetic layers is called a storage layer and able to change its magnetization direction to write data. The other magnetic layer is called a reference layer and its magnetization is fixed. The reference layer should have a large coercivity enough to not change its magnetization by writing operation and by any disturbances.

The MTJ has hysteresis as shown in Fig. 4c. The resistance of MTJ takes low value when the magnetization is in parallel configuration. That takes high value when they are in antiparallel configuration. Data 0 and 1 are assigned to each configuration.

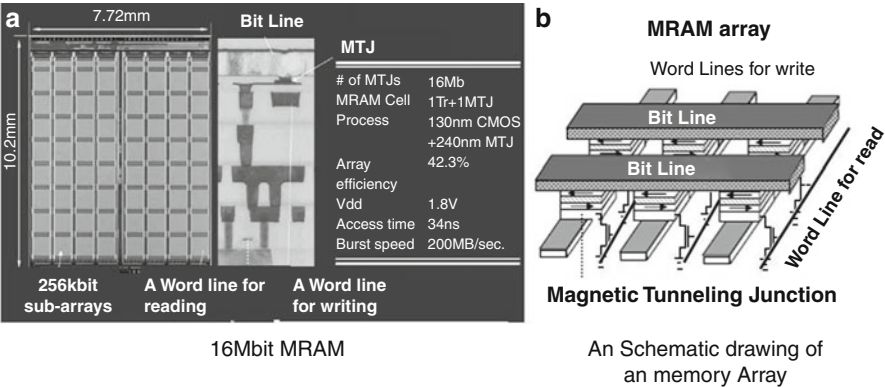


Fig. 3 16 Mbit MRAM and schematic drawing of its array [1]. (a) MRAM is a memory which uses magnetoresistance effect for reading principle. (b) Shows arrays with MTJs (Magnetic Tunnel Junctions) as storage elements

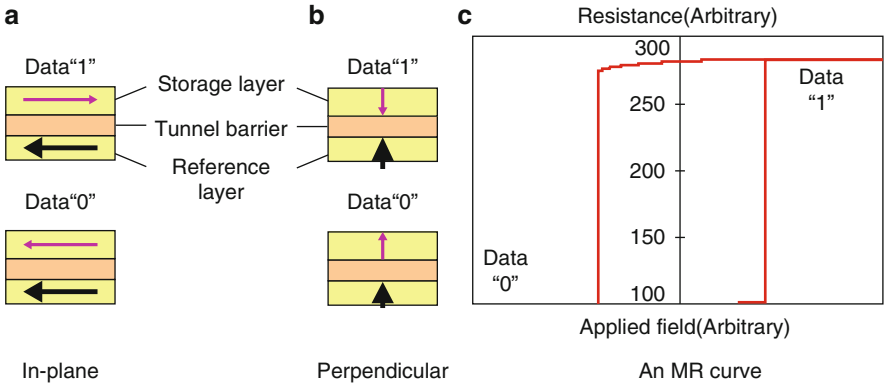


Fig. 4 Schematic drawings of MTJs and the hysteresis curve. MTJ consists of two magnetic layers and a tunneling barrier sandwiched by them. The resistance of MTJ takes low value when the magnetization takes parallel configuration and high value when they take antiparallel configuration. Data 0 and 1 are assigned to each state

Merits and Development Items of MRAM

Table 1 is a comparison table for existing and emerging solid-state memories. Before comparing those memories, let us review each memory.

DRAM (Dynamic random-access memory) is a type of a random-access memory. The unit cell consists of just a selection transistor and a capacitor and is as small as 6–8 F².

Digital data is stored in the capacitor, as either a charged state or a discharged state. Since capacitors leak charge through the transistor, the data is eventually lost unless the capacitor is charged periodically. The periodical charging is called refreshment.

The merits of DRAM are its structural simplicity (low cost) and relatively fast read and write speed. The drawback is volatility which requires the refreshment. The refreshment leads to large power consumption even in nonusage and data is lost at each shut down of power supply.

NAND flash memory is a type of storages. The unit cell consists of just one transistor with a floating gate and is as small as 4 F². Digital data is stored in the floating gate, as a charged state, half-charged state, a discharged state, etc. Since the floating gate is insulated, the charge is on hold for some period.

The merits of NAND flash memory are its structural simplicity (low cost) and nonvolatility. The drawbacks are slow writing speed and endurance limitation which is a result of trade-off with nonvolatility. Nonvolatility required perfect insulation of the floating gate. This required high writing voltage which leads to the breakdown of the insulator.

Table 1 A comparison table for existing and emerging memories. DRAMs have been used for working memories and NAND flashes for storage memories. Writing operation of MRAMs does use switching of electron’s spin and atoms do not move, while other candidates for nonvolatile memories use replacement of atoms to store information. Therefore, MRAMs are only memory which has relatively fast writing speed, unlimited endurance, and nonvolatility. This is a reason why MRAMs are thought to be best candidates for nonvolatile work memories

	Work Memory (Fast, No endurance)			Storage (High Density)		
	DRAM	MRAM	FeRAM	PRAM	ReRAM	NAND flash
Cell	Capacitor	Magnetic Tunnel Junction	Capacitor (Ferro electric)	Chalcogenide Glass	Metal/oxide /Metal	Floating Gate
Equivalent circuit						
"1" Data						
"0" Data						
Merit	1.High speed 2.Endurance 3.Low Cost	1.Non-volatile 2.Endurance 3.Speed	1.Non-volatile 2.Speed	1.Non-volatile 2.Low cost	1.Non-volatile? 2.Low cost	1.Low cost 2.Non-volatile
Demerit	1.Large stand-by current 2.Volatile	1.Large write at high density	1.Scalability limit 2.Endurance 3.Destructive read	1.Endurance 2.Large write at high density	1.Endurance 2.Large write at high density	1.Slow write 2.Endurance limit

Relocation of electrons in writing

Relocation of electrons in writing

Relocation of electrons in writing

FeRAM (Ferroelectric RAM) is categorized in a random-access memory but does not have unlimited endurance. The unit cell consists of a selection transistor and a capacitor with a ferroelectric material. The cell size is several dozens of F^2 , because the area of the capacitor should be large to have enough reading signal. Digital data is stored in a ferroelectric material, as polarization directions, upward or downward.

The merits of FeRAM are nonvolatility and moderate speed. The drawback is limitation of scalability. FeRAM also has limited endurance because polarization direction change requires relocation of atoms.

PRAM (Phase change memory, also called PCRAM, PCM, or Ovonic Memory) is categorized in a random-access memory but does not have unlimited endurance. The unit cell consists of a selection transistor or selection diode and a resistor made of chalcogenite material. The cell size is $4\text{--}8 F^2$. Digital data is stored in the resistor, as an amorphous phase, high resistance or a crystal phase, low resistance.

The merits of PRAM are nonvolatility and density. The drawback is limited endurance and large writing current at high density. The limited endurance comes from a fatigue that the cell accumulates stress in changing phases which requires relocation of atoms.

ReRAM (Resistive RAM) is categorized either in a random-access memory or in a storage depending on types of unit cells. The unit cell consists of a selection transistor or selection diode and a resistor made of dielectric material. The former cell size is $6\text{--}8\text{ F}^2$ and the latter is 4 F^2 . Digital data is stored in the resistor, as an insulation state, high resistance, or a state which has conduction paths, low resistance.

The merits of ReRAM are nonvolatility and density. The drawback is limited endurance and large writing current at high density. The limited endurance comes from a fatigue that the cell accumulates stress in changing state which requires relocation of atoms.

Merits of MRAM

MRAM has nonvolatility, relatively high read and write speed, and unlimited endurance. Especially, unlimited endurance is the merit which no other nonvolatile memories have. Unlimited endurance comes from the magnetization switching mechanism which is free from atomic relocation. HDD (Hard Disk Drive) uses magnetic materials and proved that field writing principle works at very high speed with ns pulse width and have unlimited endurance. Field-writing MRAM also proved to have relatively fast read and write speed and unlimited endurance as well [1]. But there was no evidence that STT-writing principle did work fast and have unlimited endurance. Those should be confirmed by experiments.

Experimental Proofs of Fast Switching and Unlimited Endurance

Reading principle is magnetoresistive effect, regardless of writing principle. Reading speed of STT-writing MRAMs is the same as that of field-writing MRAMs and must be relatively fast. Therefore, only switching speed of STT-writing MRAMs should be examined.

Experiments were carried out. Figure 5 shows the writing tests on perpendicular MTJs.

As shown in Fig. 5a, feeding write current with 3 ns pulse width to MTJs successfully switched the magnetization. Due to the large resistance of the MTJ, 3 ns was the shortest pulse width to feed. In order to feed shorter pulse current as short as 0.5 ns, GMR elements which had very small resistance and small RC delay were chosen as test vehicles. Writing with 0.5 ns pulse successfully switched the magnetization with 100 % probability as in Fig. 5b [10]. There is no inherent drawback in writing speed.

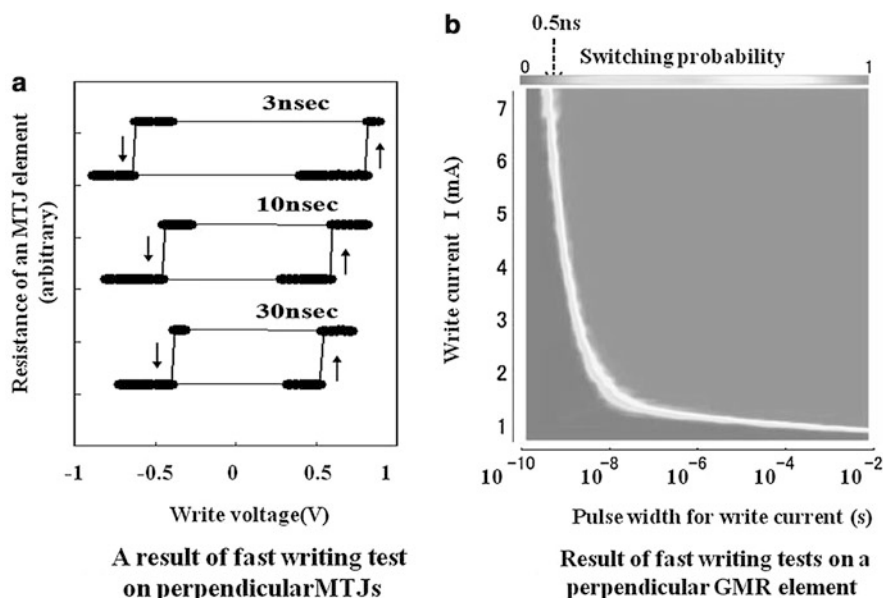


Fig. 5 Results of fast writing tests on perpendicular magnetoresistive elements [5, 10]. MTJs are shown to switch as fast as 3 ns. In order to feed short-pulse current as short as 0.5 ns, GMR elements, which had very small resistance and small RC delay, were chosen as test vehicles. As a result, writing as fast as 0.5 ns was successfully demonstrated

Next, endurance tests were carried out. The result is shown in Fig. 6. No limit up to 3.6×10^{12} was confirmed.

Those merits are thought intrinsic to MRAM because of magnetization switching mechanism. As explained at the next section, it is an electron not an atom that moves in the magnetization switching, while FeRAM, PRAM, and ReRAM need relocation of atoms in writing. This is the intrinsic reason that only MRAM does have unlimited endurance.

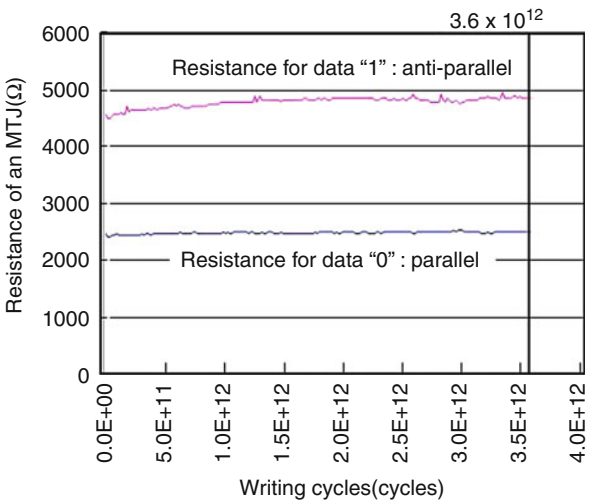
Items to Overcome

However, the field writing requires large writing current of several milliamperes. The current increases as the design node gets smaller. Therefore, the scalability is thought to be less than several hundred megabits.

In order to overcome this difficulty, STT writing has attracted much attention. But the writing current was still as large as submilliamperes which was a big obstacle to hinder the densification of STT MRAM.

Selection transistors can supply only several dozens microamperes at design nodes of 30 nm or smaller because typical drivability of current is the order of

Fig. 6 An endurance test result of an STT-MRAM. MRAMs are thought to have unlimited endurance because of its writing mechanism. In order to prove it, writing test was carried out. Even after 3.6×10^{12} writing, almost no degradation in resistance and MR were observed (Copyright Dr. Shinji Yuasa)



milliamperes per micrometer gate width. The switching current should be as small as several dozen microamperes, hopefully less than 30 μA .

Thus, the gap was in the order of 10–100.

Fundamental Principles of MRAMs

In order to understand MRAM in depth, let us study some fundamentals.

Origin of Magnetism

We know that tiny pieces of magnet have magnetization. This is because the origin of the magnet is spin of an electron as shown in Fig. 7. Each electron has not only negative charge of $-e$ but also a spin of one half. Each electron is a tiny magnet. Therefore, all we need in order to switch the magnetization direction is to just switch the spin direction of electrons and no need to relocate atoms. This is the intrinsic reason that MRAM does have fast switching and unlimited endurance.

Origin of Storage Energy (Retention Energy)

When magnetic materials are put in a room temperature environment, the thermal energy of about $300 k_B T$ agitates the magnetization to change its direction. To prevent the switching of the magnetization direction, i.e., loss of information, the magnetization direction should be strongly restrained to some directions. This restraining or storage energy per volume is called magnetic anisotropy.

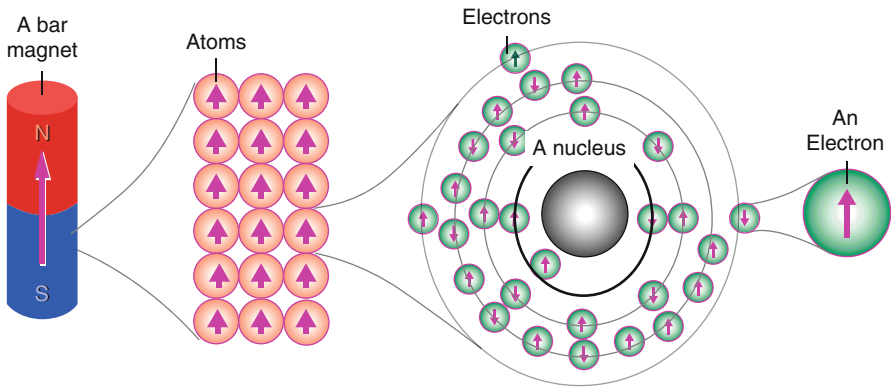


Fig. 7 The origin of magnetization. The origin of magnetization is spin of an electron. There is no need to move atoms to change the data. All we need is to change the spin direction of electrons. This gives MRAM a unique feature of unlimited endurance (Copyright Dr. Shinji Yuasa)

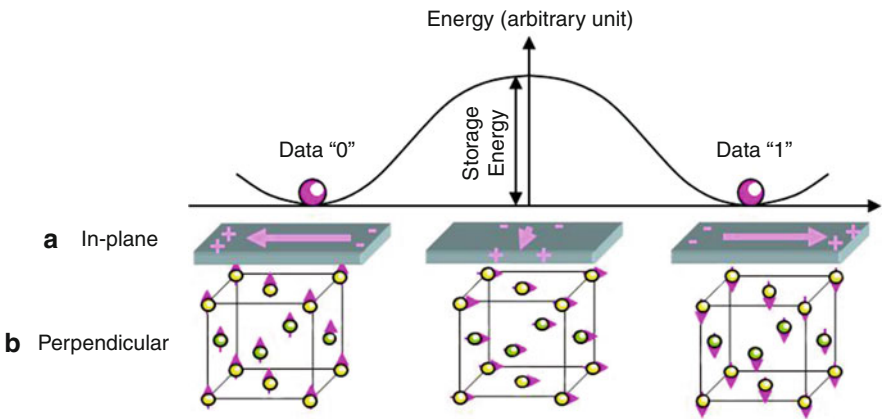


Fig. 8 Origin of nonvolatility. MTJs with in-plane magnetization use shape anisotropy for its storage energy and MTJs with perpendicular magnetization use crystal anisotropy or surface anisotropy

As explained in previous section, there are two ways to store data.

Figure 8a shows in-plane magnetization case. It uses shape anisotropy to have storage energy. Therefore, the shape of MTJ should have long axis and narrow axis.

MRAM products use MTJs with in-plane magnetization (We simply call in-plane MTJ) because the material technology are more mature than those with perpendicular magnetization (We call simply perpendicular MTJ).

However, shape anisotropy is thought not strong enough for MTJs at narrow size of 30 nm or below. This problem was overcome by perpendicular MTJs because of the below mentioned reason.

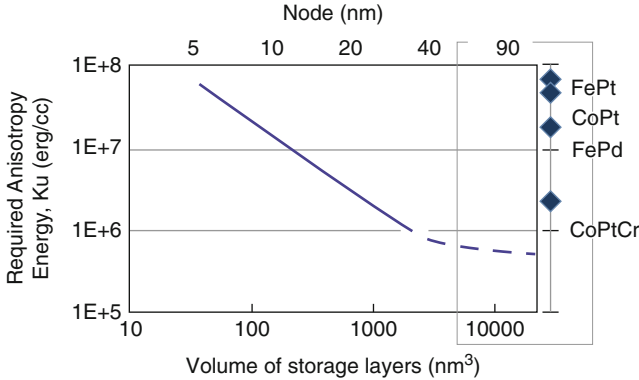


Fig. 9 Required anisotropy energy for 10-year retention (K_u). There are materials such as FePd, CoPt, and FePt with large anisotropy energy over 10^7 erg/cc which gives 10-year retention even at 10 nm node

Then, it is thought that denser MRAMs should use perpendicular MTJs in which magnetizations of the magnetic films are perpendicular to the film plane.

Figure 8b shows perpendicular magnetization case. This perpendicular magnetic anisotropy originates from the anisotropic crystal structure. The anisotropy is lattice anisotropy or interface anisotropy both of which have been well known for years.

There are perpendicular materials with large anisotropy energy densities in the order of 10^7 erg/cc, such as FePt, CoPt, and FePd as shown in Fig. 9. The cell can have nonvolatility even at 10 nm node. This is one of the reasons why it is thought that perpendicular magnetic materials must be used for future dense MRAMs.

Reading Principle

The present MRAMs use TMR effect as reading principle. Figure 10 shows mechanism of TMR.

Electrical current flowing inside nonmagnetic materials has equal numbers of up-spin and down-spin electrons. But that flowing inside magnetic materials has different numbers of up-spin and down-spin electrons. This unbalance is called spin polarization P , which is defined as $P = [n(+)-n(-)]/[n(+)+n(-)]$, where $n(+)$ and $n(-)$ are the numbers of up-spin and down-spin electrons, respectively.

Figure 10a explains tunneling current conduction when the magnetization of a storage layer and that of a reference layer gets parallel. The areas with blue color show the densities of states for electrons having up-spin and those with pink color show those having down-spin. The electrons at the highest energy level, i.e., Fermi level mainly travel between the layers and the tunneling current conducts. Since the two layers have the same density of states, every electron at Fermi level can travel between them, i.e., the tunneling probability is large and the MTJ shows low resistance.

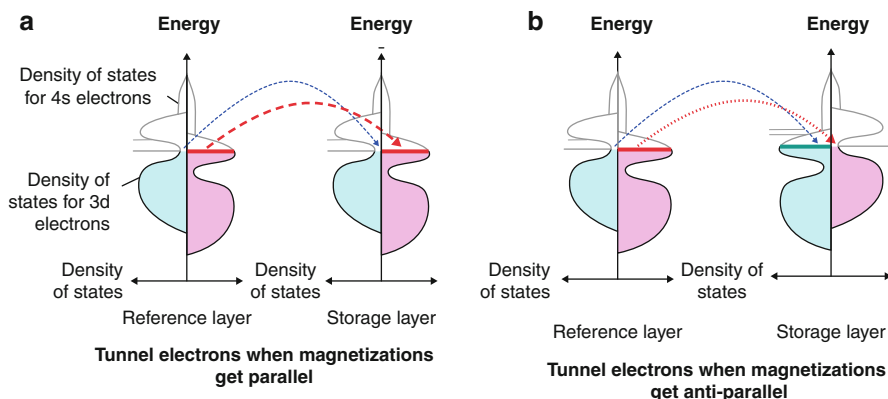


Fig. 10 Mechanism of resistance change in MTJs. The probability of the electron tunneling is proportional to the density of states at Fermi level of a reference layer and that in a storage layer. The probability takes maximum when magnetization in a reference layer and that in a storage layer takes parallel states and minimum when those take antiparallel state

Figure 10b explains tunneling current conduction when magnetization gets antiparallel. Since the two layers have different density of states at Fermi level, the tunneling probability is limited by the smaller density of states at Fermi level and the MTJ shows large resistance.

Magnetoresistance (MR) is defined as $[R(AP)-R(P)]/R(P)$, where $R(P)$ and $R(AP)$ are the resistances of the parallel and antiparallel states, respectively, and is related to P , spin polarization, by $MR = 2P^2/(1 - P^2)$. For ideal magnetic materials with $P = 100\%$, MR should be infinite. In most magnetic materials, there are down-spin electrons as well as up-spin electrons and P is much smaller than 100% . MR becomes finite.

At first, an amorphous Al_2O_3 tunnel barrier was used for MTJs. The theoretical limit of MR for $CoFe/(Amorphous\ Al_2O_3)/CoFe$ is only 80% because P for $CoFe$ alloy is just 55% .

Since $\Delta R = R(AP)-R(P)$ should be larger than the variation of MTJ's resistance in a MRAM chip as shown in Fig. 11, over 150% of MR is required for gigabits density.

Tunnel barrier material was changed from AlO amorphous material to MgO crystal to meet high MR specification.

In a body-centered-cubic (bcc) magnetic material, electrons with three different quantum mechanical wave functions of Δ_1 , Δ_2 , and Δ_5 coexist at the Fermi. Electrons with the Δ_1 are all up-spin, i.e., $P = 100\%$. On the contrary, electrons with the Δ_2 and Δ_5 have both up- and down-spins. In MTJs with AlO amorphous material, electrons with all wave functions tunnel as shown in Fig. 12a. Therefore, averaged P becomes about 50% and MR falls to less than 100% .

MgO (100) with a NaCl crystal structure allows only electrons with the Δ_1 wave function to tunnel between the bcc-magnetic layers(100) while it blocks electrons

Fig. 11 Required MR. MR should be large enough to separate the data “0” bits and data “1” bits. As the density increases, larger MR is required

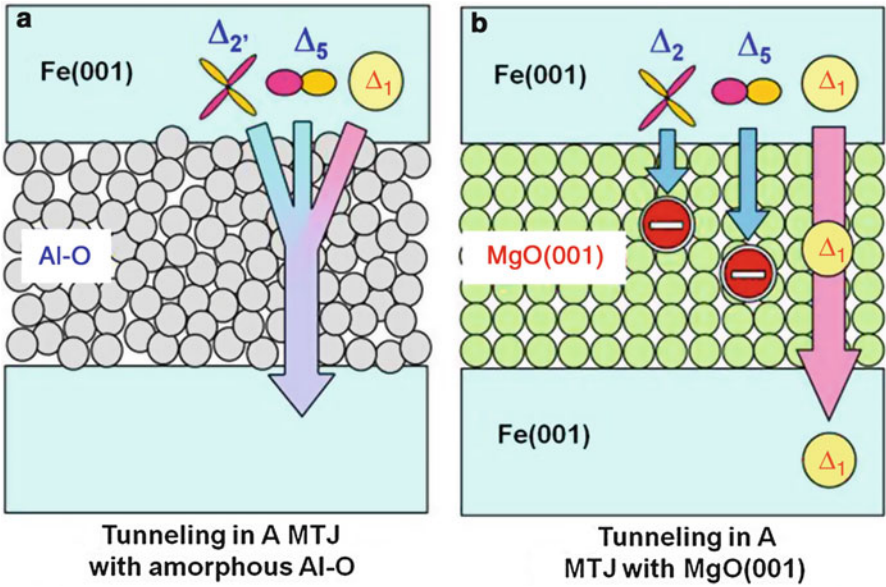
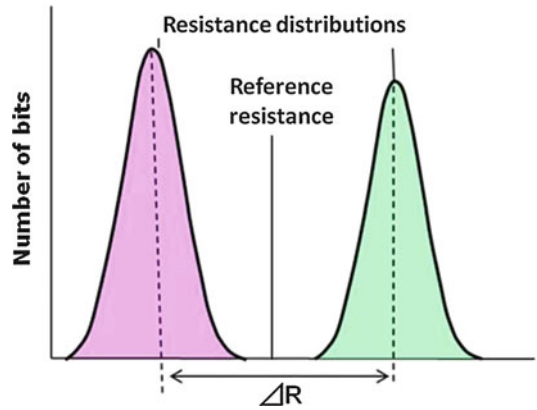


Fig. 12 Mechanism of resistance change in MTJs with MgO tunnel barrier

with the Δ_2 and Δ_5 wave function level as shown in Fig. 12b. This filtering effect was proved to increase MR over 500 % for a simple Ta/CoFeB/MgO/CoFeB/Ta structure.

Figure 13 shows a (001) oriented Fe/MgO/Fe.

MTJs which used in MRAMs are required to have stable operation and the structure needs additional magnetic layers to fix a reference layer and shift-cancelation layer to adjust the center of the hystericis curve. These additional layers usually degrade MR to less than 300 % because of diffusion of the materials.

Fig. 13 MTJ film with MgO tunnel barrier grown on MgO substrate [22]. (001) Orientated growth is a key to large MR. Large MR was demonstrated by single crystal Fe/MgO/Fe

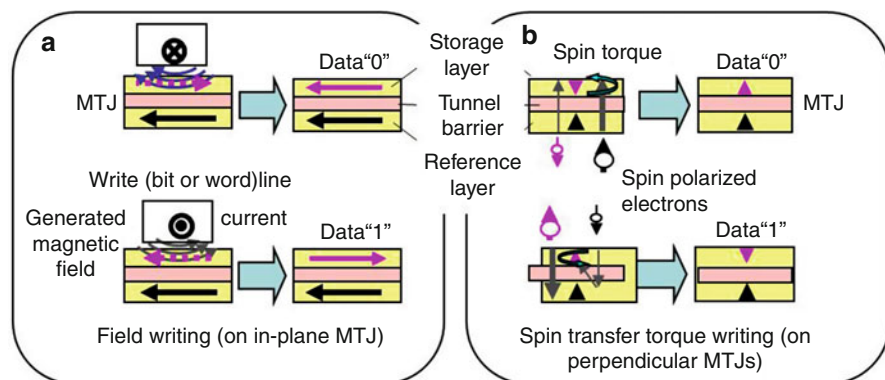
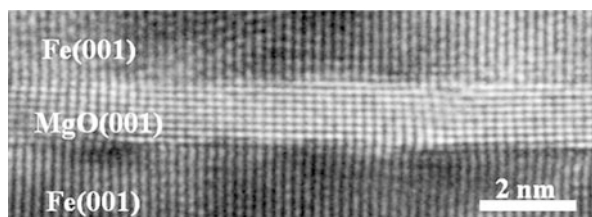


Fig. 14 Writing principles for MRAMs. (a) Magnetic field generated by electrical current switches magnetization in a storage layer. (b) Spin torque transferred by electrons switches magnetization in a storage layer

Writing Principle

There are several writing principles. Field-writing and STT-writing are well studied and promising principles for MRAM application.

Field-writing is old principle and used in commercialized MRAMs. Magnetic field generated by electrical current switches magnetizations as shown in Fig. 14a.

STT-writing is relatively new principle and tried to apply to future dense MRAMs.

In 1996, two theoreticians Slonczewski (1996) and Berger (1996) independently proposed a method called current induced magnetization reversal for changing the magnetization direction [11], [12]. This is addressed as STT- writing now.

The STT-writing is shown in Fig. 14b.

Because conduction electrons in magnetic materials are spin-polarized, the tunneling current carries the spin angular momentum to apply torque to the magnetization of the storage layer. This transfer of angular momentum exerts the torque to the magnetization to switch.

For antiparallel to parallel switching, the upper figure, current has more number of electrons with up-spin than those with down-spin. The magnetization of the storage layer switches to upward.

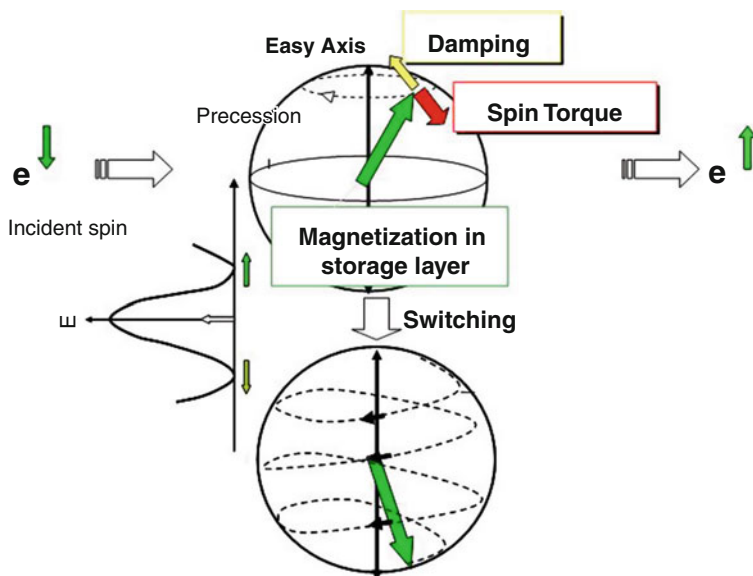


Fig. 15 Schematic explanation of a principle of spin transfer torque writing. The magnetization in a storage layer precesses around its easy-axis. Electrons with opposite spins interact with the magnetization to apply spin torque. But the damping torque keeps the magnetization from switching. More electrons are fed to enlarge the spin torque. Once the spin torque exceeds the damping torque, a radius of the precession gets larger and the magnetization flips

For parallel to antiparallel switching, the bottom figure, electrons with down-spin cannot tunnel, because the corresponding density of states in the reference layer is small. As a result, electrons with down-spin are accumulated in the storage layer to switch the magnetization downward.

Figure 15 shows details of STT-writing, spin of the incident electron applies spin-torque to the magnetization of the storage-layer but the damping torque prevents it from switching. Therefore, more polarized electrons are needed to switch the magnetization to overcome the damping.

History of Developments on MRAM Technologies

The first magnetic memory which used magnetic materials is a core memory developed in 1955. The core memory was fast and cheap enough to use as a cache memory as well as a storage at that time. It was actually a universal memory. This is shown in Fig. 16. It is surprising that all of the memory and the storage in the 1960s were nonvolatile and made with magnetic materials. The core memory used magnetic field generated by electric currents to write data and electromagnetic induction to read data, those were found in 1820 by H.C. Oersted and in 1831 by M. Faraday, respectively as shown in Table 2.

Fig. 16 Memory-storage hierarchy in the 1960s. In the 1960s, magnetic core memories were used as universal nonvolatile memories which worked

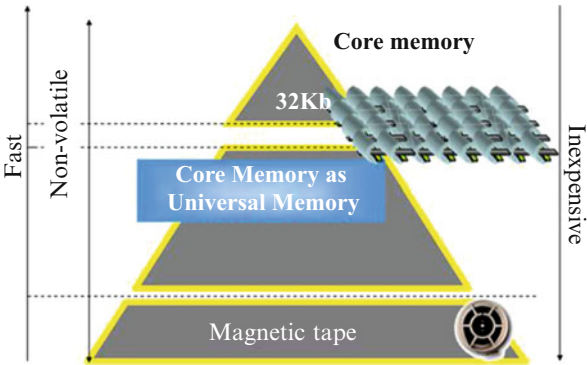


Table 2 History of developments on magnetic memories. Several innovations have been made to give birth to magnetic memories. Room temperature TMR, spin transfer torque, MgO tunnel barrier, and MTJs with perpendicular magnetization may lead to a birth of gigabits MRAMs

Break throughes		Developments for memory applications		
Year	Breakthrough technologies	Year	Developed memories	Comments
1820	Magnetic field generation by electric currents (H.C. Oersted)	1955	Core Memory	Used as universal memory
1831	Electromagnetic induction (M. Faraday)			
1856	Magnetoresistance in Fe (Lord Kelvin)			
1988	Giant Magneto-Resistance (GMR) (A. Fert and P. Grünberg)	1980s	Field-writing MRAM (AMR)	–
1994	Room temperature Tunnel Magneto-Resistance (TMR) (T. Miyazaki and J.S Moodera)	1990s	Field-writing MRAM (GMR)	–
1996	Spin Transfer Torque (STT) (J. C. Slonczewski and L. Berger)	1990s	Field-writing MRAM (TMR)	DARPA project (1996–)
2004	TMR with MgO Tunnel barrier (S. Yuasa and S. Parkin)			
2006	STT-writing with Perpendicular GMR (S. Mangin et al.)			
2007	STT-writing with Perpendicular TMR (H. Yoda et al.)	2000s	STT-writing MRAM (in-plane)	–
		2000s	STT-writing MRAM (perpendicular)	NEDO project (2006–)

In the 1980s, field-writing MRAM had been developed. It used magnetic field generated by electric current to write data and AMR (Anisotropic Magneto Resistance effect) to read data which was found in 1856 by Lord Kelvin. This is the first MRAM (Magnetoresistive Random Access Memory).

In the 1980s, reading principle was changed to GMR (Giant MagnetoResistance effect) which was found in 1988 by A. Fert and P. Grunberg [13], [14]. They discovered the GMR effect in multilayer structures of few-nm-thick ferromagnetic and nonmagnetic metals. The resistivity of the multilayer depends on the relative

direction of magnetizations of ferromagnetic layers. The magneto-resistance was 85 % at 4.2 K and 1.5 % at room temperature.

In the mid 1990s, reading principle was changed to TMR (room temperature Tunnel MagnetoResistance effect) which was found in 1994 by T. Miyazaki and Moodera [15], [16].

They used amorphous Al_2O_3 for the tunnel barrier. Due to the higher MR effect and large resistance of tunneling barrier, MTJs contributed to increase the capacity of MRAM to megabits level. After intensive research and development such as DARPA MRAM project, MRAM was commercialized in 2003.

There remained a big obstacle to prevent an increase in memory density or capacity of MRAM beyond gigabits. It was a large writing current. While MRAM used the quantum mechanical effect of TMR as a reading principle, MRAM still relied on a traditional writing principle, i.e., field generated by write current the same as the core memory used. Because the efficiency of the field generation was small, the writing current was the order of milliamperes. Since narrow write lines cannot flow that large current, memory density was never beyond 256 megabits.

In the 2000s, writing principle was changed to STT-writing (Spin Transfer Torque) which was proposed as CIMR (Current Induced Magnetization Reversal) by J.C. Slonczewski and L. Berger in 1996 [11], [12].

Experimental demonstrations of the STT-writing were conducted by Myers et al. (1999) with GMR, Huai et al. (2004) with Al_2O_3 -MTJ, and Kubota et al. (2005) with CoFeB/MgO/CoFeB-MTJ, respectively [17–19].

Since STT-writing use an interaction between spin of electrons which locate very close to each other, it is expected very efficient way of switching magnetizations. However, the writing current was still the order of milliamperes. The biggest obstacle of large write current had never been solved. We still had to wait for the final breakthrough.

MTJ tunnel barrier material was changed from AlO amorphous material to MgO crystal material proposed by, Butler et al. (2001) and Mathon et al. in 2001 and experimentally demonstrated by S. Yuasa and S. Parkin [20–23]. As a result, MR increased from 70 % to over 200 % for MTJs with in-plane magnetization which had stable reference layer and were applicable to memories. The MgO tunnel barrier technology solved a poor reading signal problem for gigabit density. Although over 500 % of MR have been demonstrated but the elements did not have stable reference layer which degraded MR because of diffusion of the element to the barrier. The MTJs without stable reference-layer cannot be applicable to memories nor to HDD heads. Largest MR actually available for those device applications is limited to about 250 %.

In April of 2006, a development plan of perpendicular MTJs to realize gigabits MRAM was submitted by Toshiba, AIST, Miyazaki-lab. of Tohoku university, Suzuki-lab. of Osaka university, Nakatani-lab. of the University of Electro-Communications, and NEDO (New Energy and Industrial Development Organization) with an experimental evidence of STT switching using perpendicular MTJs and accepted as its project. This was the start of challenging to dense MRAMs.

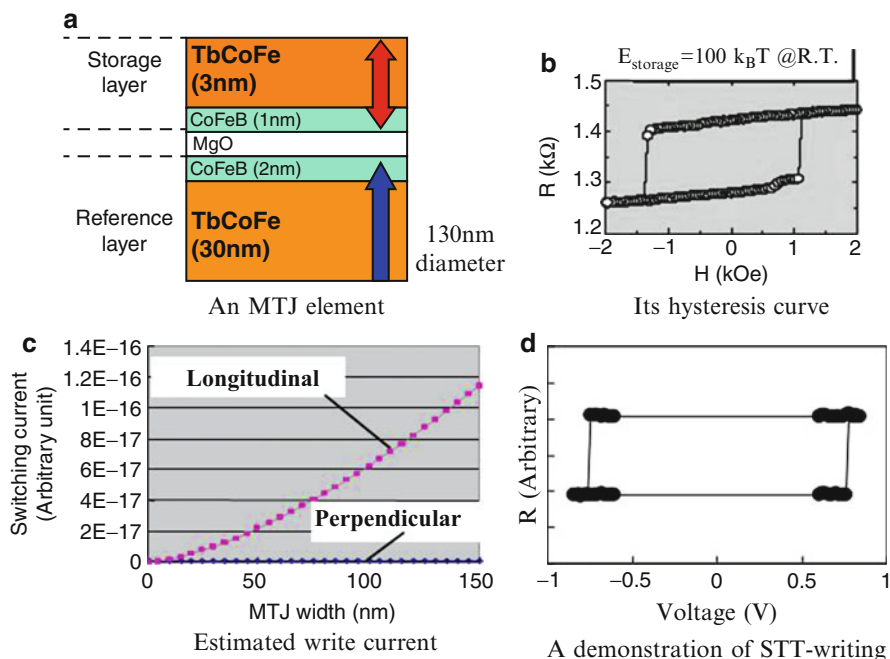


Fig. 17 Proposal of perpendicular MTJs for STT-writing and its switching demonstration [2], [3]. (a, b) Data which show that MgO/CoFeB/TbCoFe have perpendicular anisotropy and magnetoresistance effect. (c) Simulation results which show a perpendicular MTJ have smaller switching current regardless of MTJ size. (d) First demonstration of STT-switching using the perpendicular MTJ

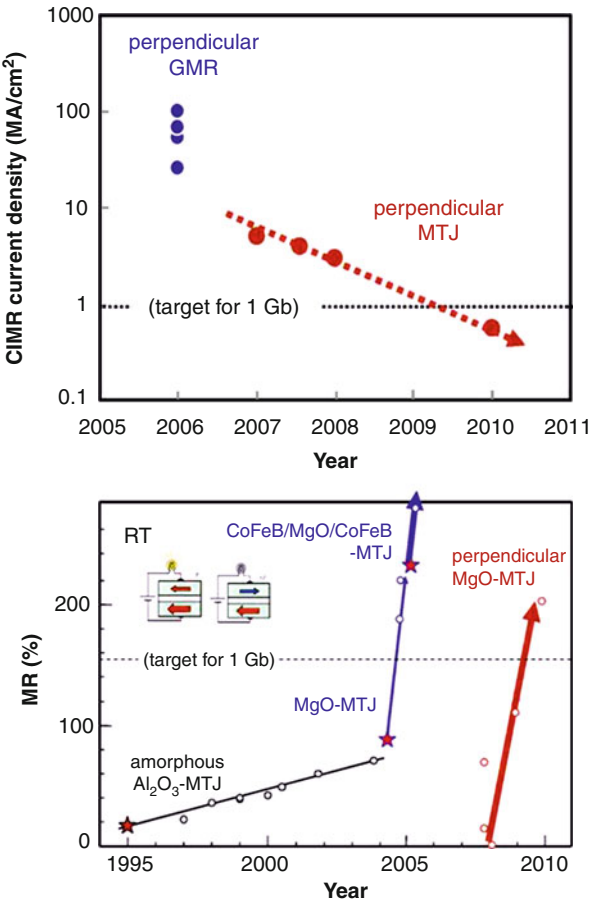
In June of 2006, challenging developments on perpendicular MTJs was started as NEDO's spintronics nonvolatile devices project.

At that time, results of STT writing on perpendicular GMR were already reported (Meng, 2006; Mangin, 2006; Seki, 2006) [24–26]. But the MR were as small as about 1 % and the write current density was as large as 26–100 MA/cm², i.e., the write current were the order of milliamperes or larger. Increasing MR and reducing the write current were a quite difficult challenge. Even growing perpendicular MTJs was thought almost impossible.

In 2007, STT-writing on perpendicular MTJ was disclosed to public by H. Yoda et al. which opened a way to gigabits density MRAM as shown in Fig. 17 [2]. In the demonstration, perpendicular MTJ made of perpendicular CoFeB/MgO/CoFeB stack was used shown in Fig. 17a. TbFeCo layer was deposited on the CoFeB layers to increase their storage energy. As a result, large storage energy of 100 k_BT was realized as shown in Fig. 17b. Simulation results were shown in Fig. 17c which predicted a perpendicular MTJ has smaller switching current regardless of MTJ size. The world's first demonstration of STT-writing on perpendicular MTJs was demonstrated as shown in Fig. 18d.

Figure 18a shows a reduction trend of switching current density, J_s .

Fig. 18 Advancements in MR and Jc. Perpendicular MTJs achieved large MR of over 200 % and small Jc smaller than 1 MA/cm² which were targets for Gbit density MRAM



Finally, J_s was successfully reduced to 0.3–0.5 MA per cm², which corresponded to write current of just 7–9 μ A during the project.

This was the final breakthrough for gigabits density STT-writing MRAM, fast and dense nonvolatile random access memories. Of course, perpendicular MTJs was the key. Every data was demonstrated by the NEDO project team and every demonstration used perpendicular MTJs made of perpendicular CoFeB/MgO/perpendicular CoFeB. Additional perpendicular layer such as FePd or CoPt were deposited on the CoFeB layers just to increase their storage energy.

Figure 18b shows an increase in the trend of MR. MR was successfully increased to over 200 %. Again every data was demonstrated by the NEDO project team.

By using the outcome of spintronics nonvolatile devices projects, Toshiba demonstrated the integration of 64 megabits STT writing MRAM in 2010 shown in Fig. 19 [27].

NEDO's spintronics nonvolatile devices project played a big role for MRAM history.

Fig. 19 64 Mbit STT MRAM [27]. Perpendicular MTJs were incorporated to reduce the cell size

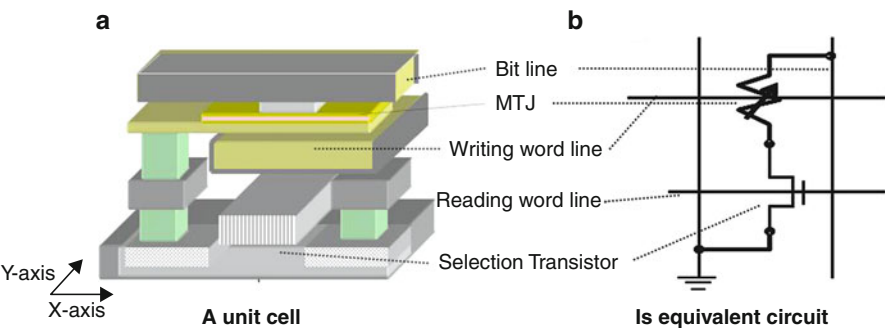
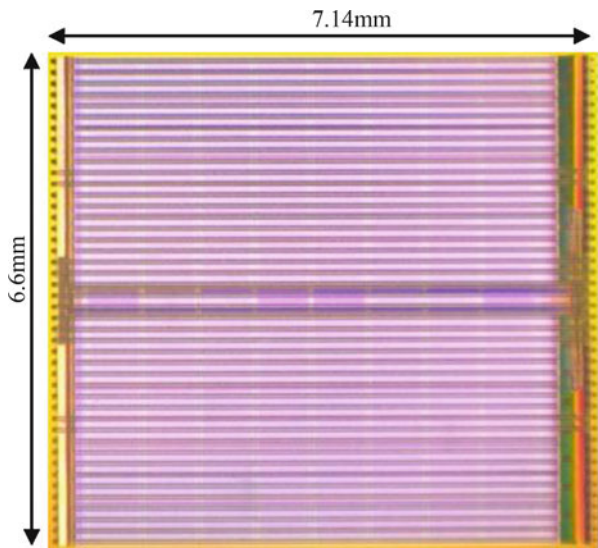


Fig. 20 Schematic drawing of a unit cell for field writing MRAMs and its equivalent circuit. A unit cell consists of an MTJ and a selection transistor. The transistor is turned on only in reading to select a cell. In writing, the bit line and the writing word line are turned on to flow currents to generate magnetic field. Switching threshold and currents are set so that only MTJ sits at the cross point switches

Field-Writing MRAM

Unit Cell

Figure 20 shows a schematic drawing of a unit cell of the field-writing MRAM and its equivalent circuit. It consists of a selection transistor, an MTJ, a bit-line, and two word-lines, one for reading the other for writing. The selection transistor is used in reading the data to feed a sense(reading) current to the MTJ by the bit-line and the read word-line. In writing, write currents are fed to the bit-line and the write word-line.

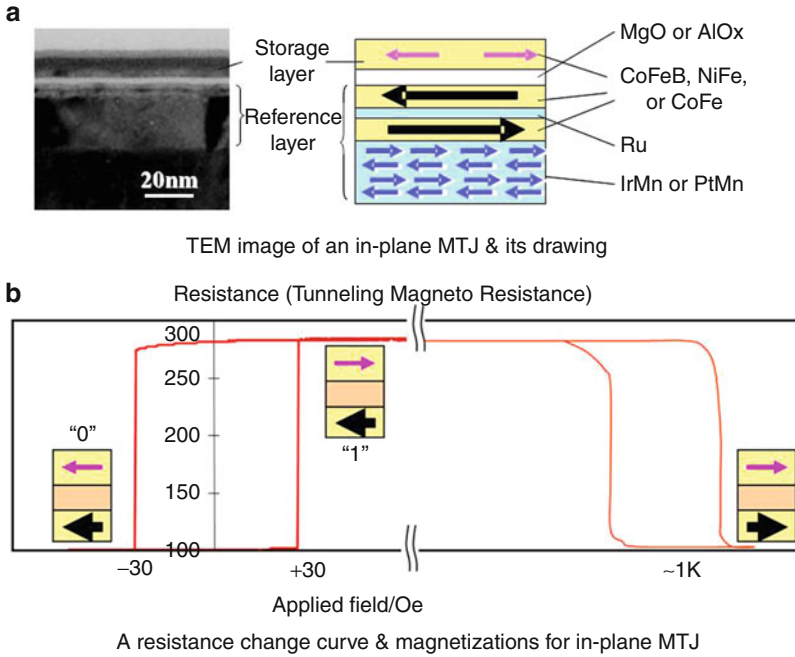


Fig. 21 An in-plane MTJ. CoFeB/Ru/CoFeB synthetic antiferro reference layer center the storage-layer's hysteresis curve. The antiferromagnetic layer made of IrMn or PtMn fixes the reference layer magnetization

The plane shape of the MTJ is anisotropic and not simple but unique to control the switching field distribution. This makes the cell size bigger than $8 F^2$.

Figure 21 shows MTJ material used for field-writing MRAMs and STT-MRAMs.

The magnetization in the storage layer should take a rightward direction and leftward direction. While that in reference layer should take just one direction i.e., should be fixed. In order to realize them, antiferro magnetic layer is placed on the reference layer to stabilize its magnetization. Further, CoFeB/Ru/CoFeB synthetic antiferro layer is used as the reference layer to center the storage-layer's hysteresis curve.

This complicated stack has to pay a penalty for the interdiffusion problem when it is annealed. This limits MR around 250 %.

Issues of Field Writing MRAM

There were two major issues to field writing MRAM, large write current and so-called write-disturbance of half-selected bits.

As stated before, field generation by current is not efficient and the large current is required to generate required field.

Fig. 22 An astroid curve and writing fields

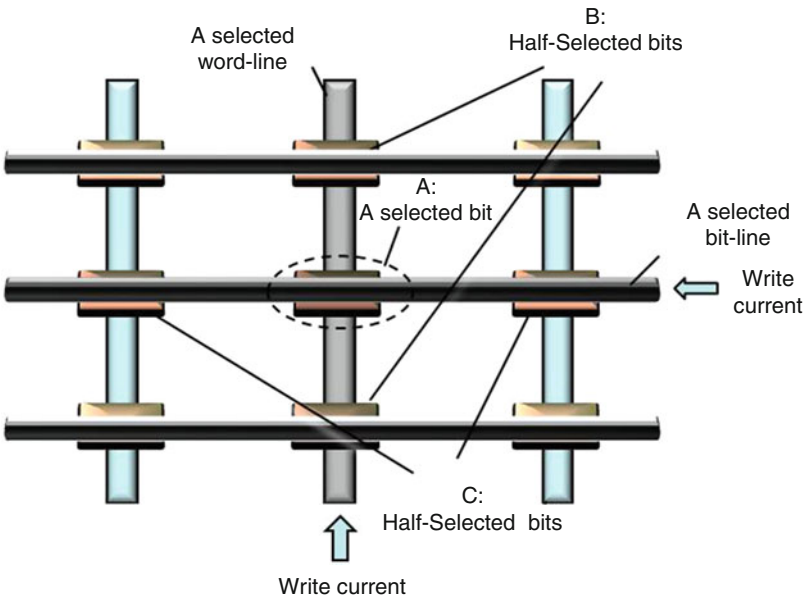
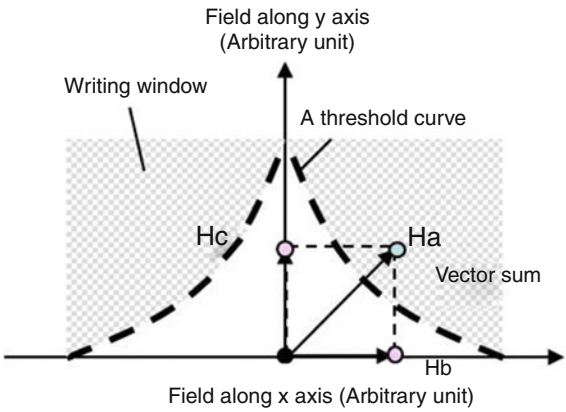


Fig. 23 Selection of write bit. The bit-line and the word-line are activated to feed write currents. At the cross-point, there is the selected bit to that the vector sum field is applied. Beneath the selected bit-line and over the selected word line, there are half-selected bits to that only x-axis field or y-axis field are applied. If the field is close to the astroid curve or larger than that, miss- writing occurs. This is called disturbance of half-selected bits

Bit selection in reading has no problem because the selection transistor can select the bit to read. But selection in writing is troublesome. Figures 22 and 23 show how to select bits to write.

A magnetic material with one-axis anisotropy has the threshold curve called astroid curve as shown in Fig. 22. Write filed of H_a is a vector sum of bit line field of

H_b and word line field of H_c. The threshold curve is designed such that H_a exceeds the curve but H_b and H_c do not in order to write only the selected bit of A and not to write the half selected bits of B and C.

But the asteroid curves vary from bit to bit. If there are some bits which have smaller asteroid curves than H_b or H_c, miss writing occurs. This is called write disturbance of half-selected bits that had hindered the development of field writing MRAMs.

Key Technologies for Field Writing MRAM

Cladding Wires (Yoke Wires) to Ease Large Write Current

As shown in Fig. 20, both the bit-line and word-line have claddings on those three faces made of magnetic layers. The wire is called as cladding wire or yoke wire [1, 28]. This wire is a nano electro-magnet that enhances the magnetic field about twice. But the write current is still the order of milli-amperes.

Write-Disturbance Robust MTJ Design

Savchenco discovered toggle-writing (Savchenco switching) to solve the write disturbance of half-selected bits [29].

In this technology, synthetic antiferro storage layer is used that composed of NiFe/Ru/NiFe. The magnetizations couple antiferromagnetically and favor the antiparallel configuration as shown in Fig. 24a, c. The magnetizations take scissoring configuration when applied field exceeds the threshold curve shown in Fig. 24b but do not switch.

In order to switch the magnetization, it is necessary to apply the rotational field.

In this Savchenco switching scheme, the easy-axis, i.e., length direction of the MTJ is placed 45° away from x-axis. First, field along y-axis larger than the threshold curve is applied. Second, x-axis field larger than the threshold curve is added. Third, y-axis field is turned off. Finally, x-axis field is turned off. In this manner, the applied field rotates as shown in Fig. 24c. Since the magnetizations take scissoring configuration shown in the figure, this scissoring configuration follows the rotation of the applied field to switch.

The beauty of this technology is write disturbance robustness. The half selection field, i.e., x-axis field or y-axis field cannot switch the magnetizations. Only the rotational field can.

This technology solved the write disturbance of half-selected bits.

STT-Writing MRAM

Even though the cladding wires enhanced the magnetic field about twice, the write current was still as large as the order of milliamperes. It is quite difficult to shrink the design rule of field-writing MRAMs, because wires with small cross-section cannot feed the large current. Field-writing MRAMs are thought to lack in scalability beyond 100 megabits.

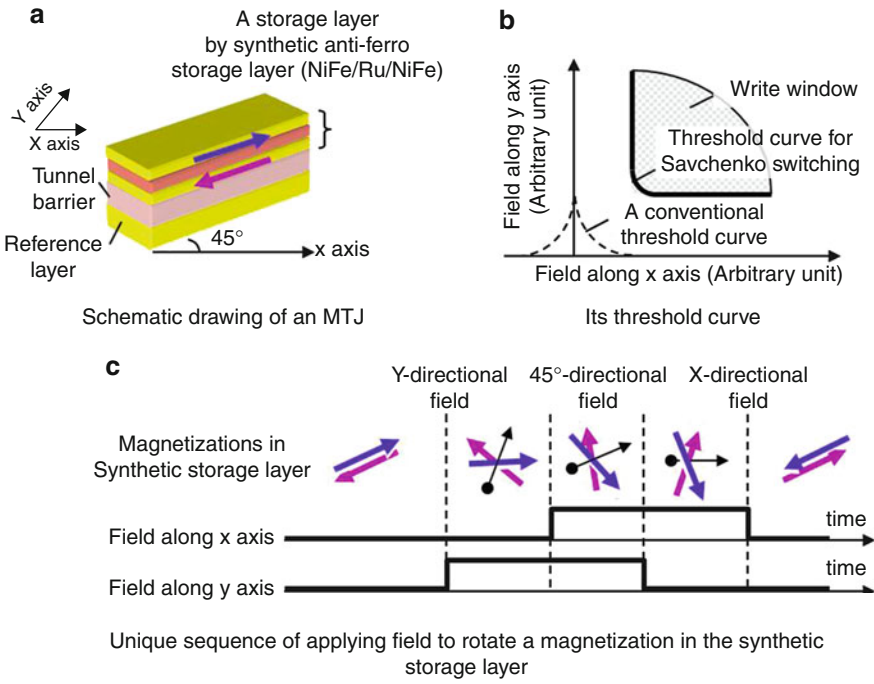


Fig. 24 Savchenko switching (Write disturb Robust Technology) [29]. In this technology, bit-lines and words are activated so that the vector sum field rotates as indicated as black arrows. The magnetizations of the synthetic antiferro storage layer take the scissoring directions as shown and follow the field rotation to switch

This is a reason why STT-writing MRAMs have been developed intensively.